

DemandIQ

End-to-End Sales Forecasting & Demand Intelligence System

Anish Reddy | Data Science Intern Project

16.6%

Forecast MAPE

Weighted ensemble of 4 models

+29.9%

vs. Seasonal Baseline

Improvement over naive benchmark

8

Dashboard Pages

Live, interactive Streamlit app

793

Customers Segmented

RFM clustering, 4 segments

90%

Conformal Coverage

Model-agnostic prediction intervals

102

Notebook Cells

Zero execution errors, end-to-end

What's Inside

- 4 forecasting models benchmarked against 2 naive baselines (SARIMA, Prophet w/ holidays, XGBoost, weighted Ensemble)
- SHAP explainability + model-agnostic conformal prediction intervals (distribution-free uncertainty bounds)
- Dual-method anomaly detection AND K-Means clustering applied to both products and customers (RFM)
- Monte Carlo inventory cost simulator (newsvendor-style) + an A/B testing lab with power analysis
- Auto-generated executive briefing with one-click PDF export, live from the current data



Scan for the GitHub repo
(replace with your live
Streamlit app link)

Tech Stack

Python * Pandas/NumPy * XGBoost * Prophet * Statsmodels (SARIMA) * SHAP * scikit-learn * SciPy * Streamlit * Plotly